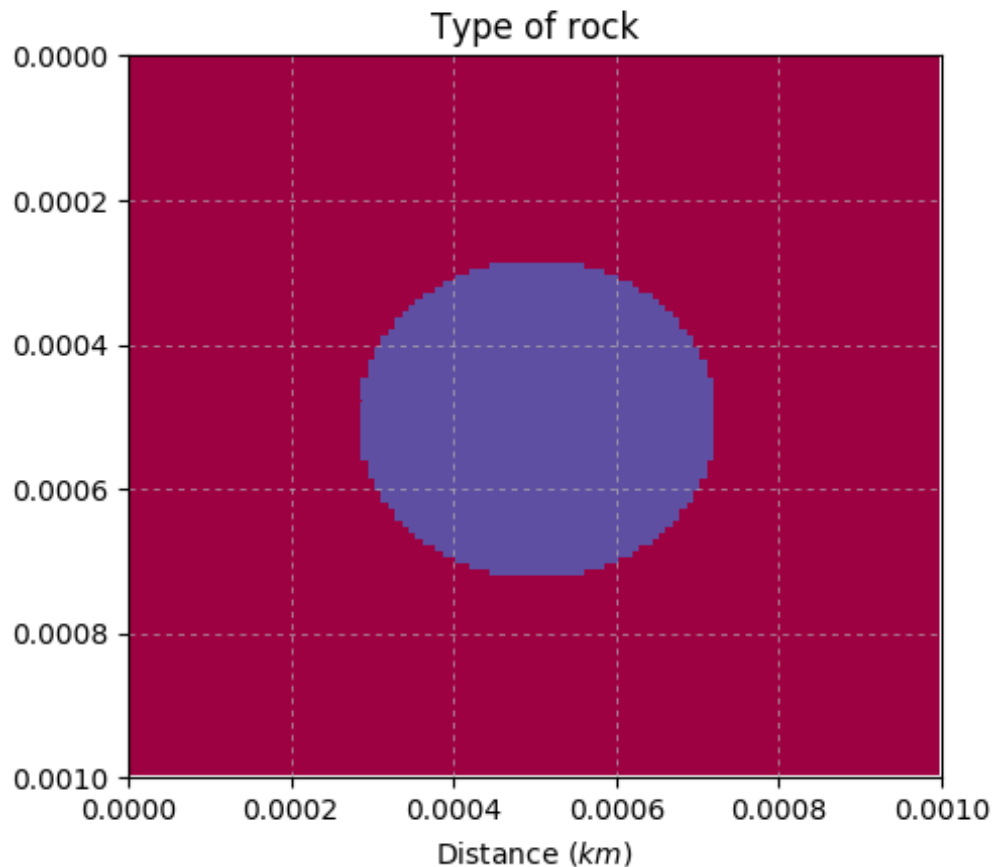


Simple example

benchmark



All boundary conditions are set to no slip.

Square box 1x1

There is a circle centered at $x=0.5$ $y=0.5$
and radius 0.22

```
if (block)
```

```
    rockm(jm,im)=1; % rock type
```

```
    ro(jm,im)=1; % density
```

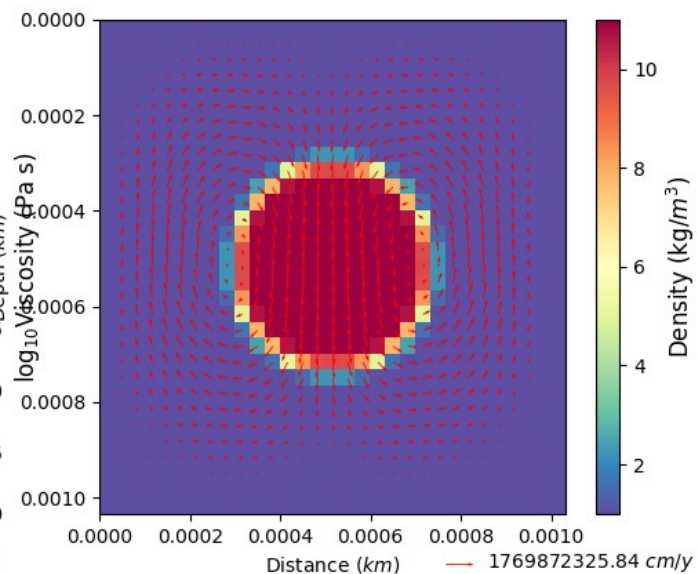
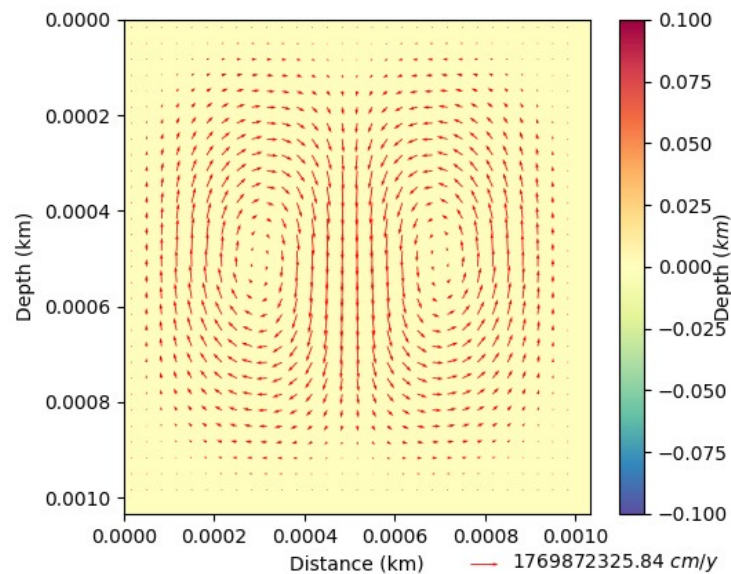
```
    nu(jm,im)=1; % viscosity
```

```
if (circle)
```

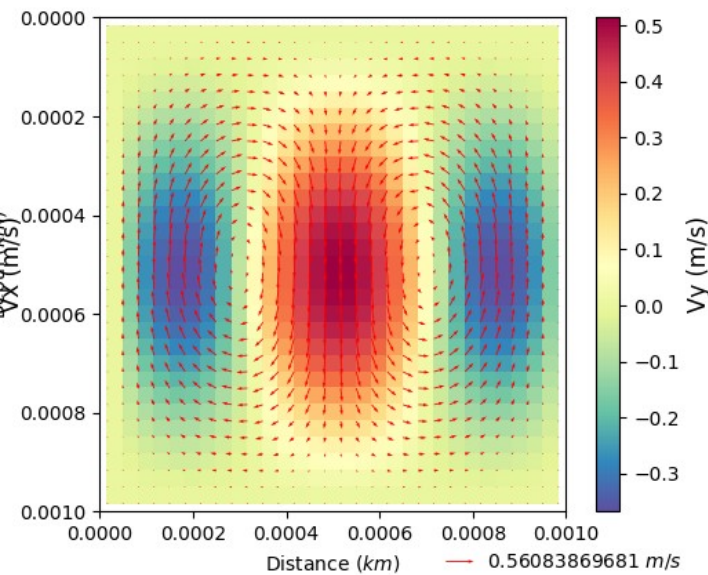
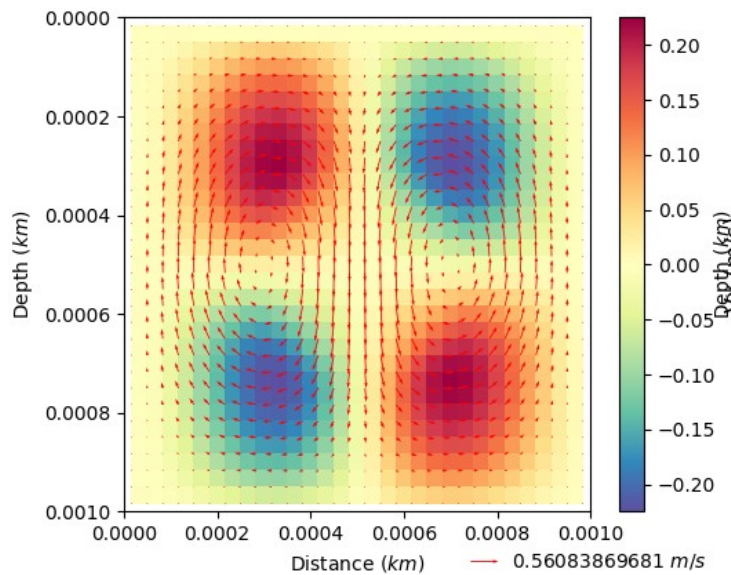
```
    rockm(jm,im)=2;
```

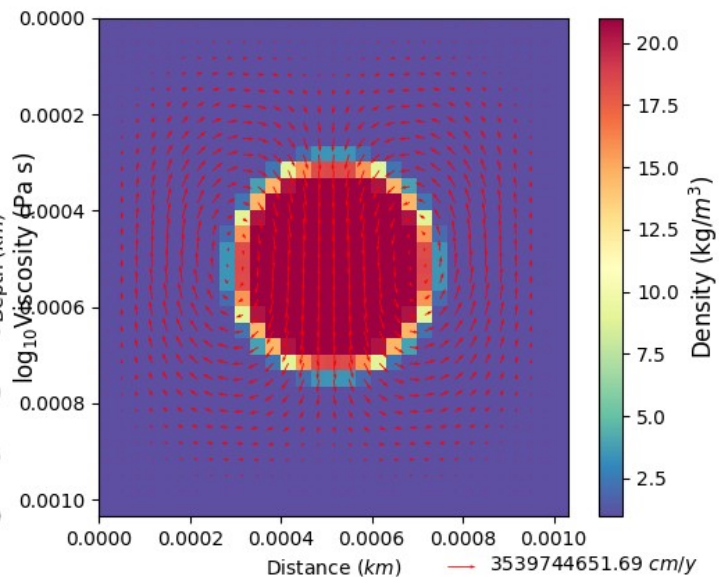
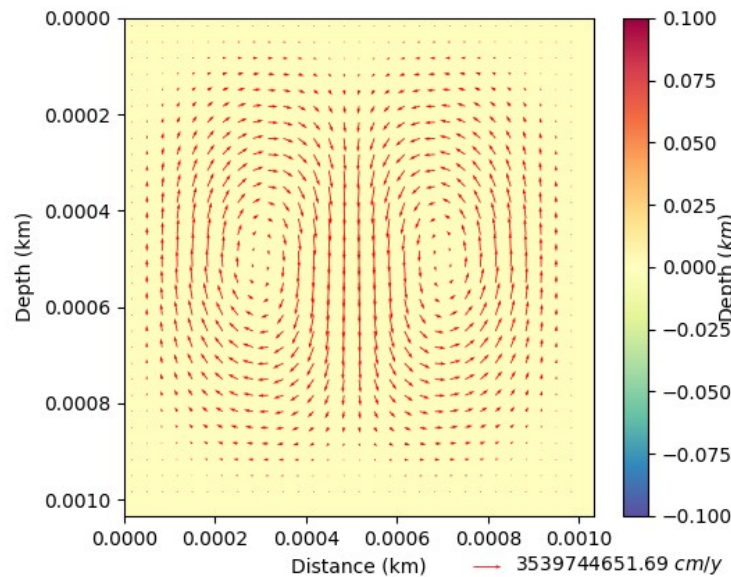
```
    ro(jm,im)=11 or 21;
```

```
    nu(jm,im)=1 or 1e6;
```



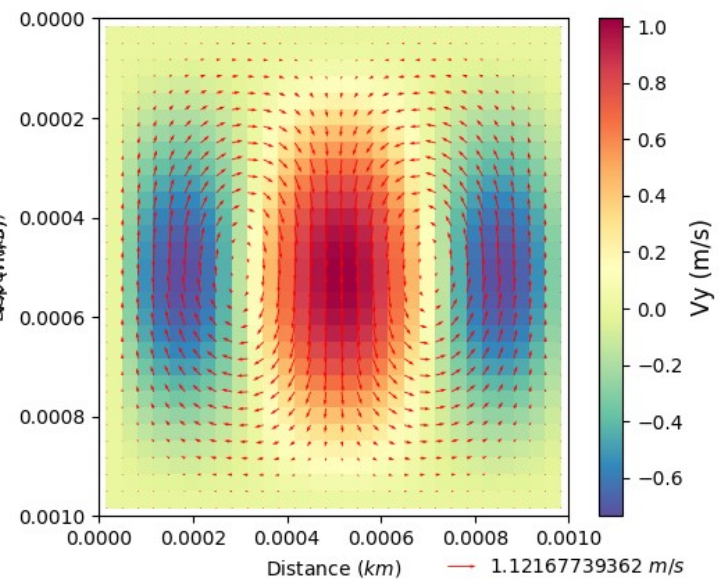
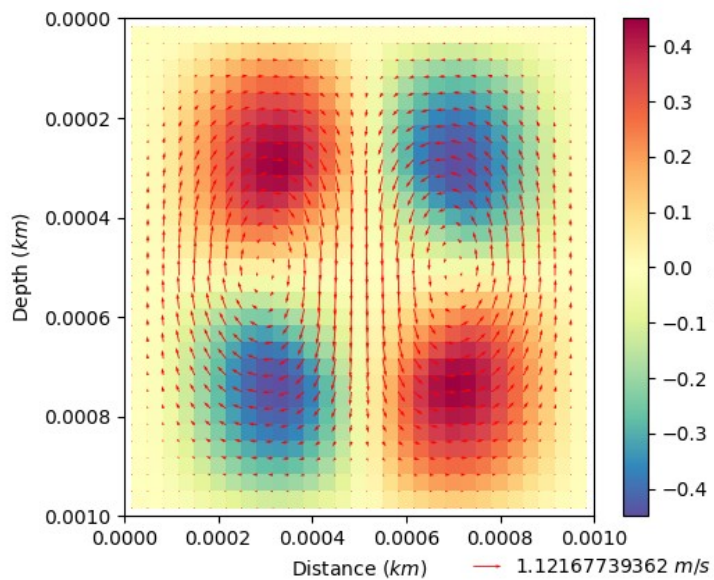
In circle:
Density= 11
Viscosity= 1

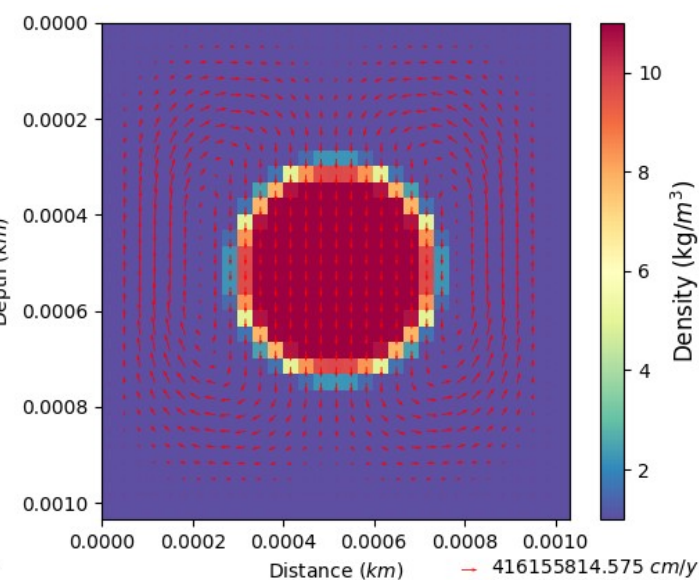
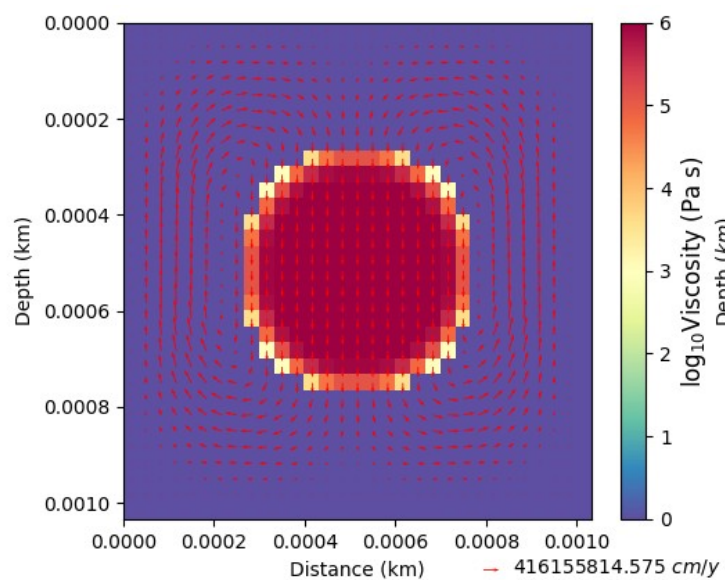




Only change the density.

In circle:
Density= 21
Viscosity= 1





Only change the Viscosity.

In circle:
Density= 11
Viscosity= $1e6$

